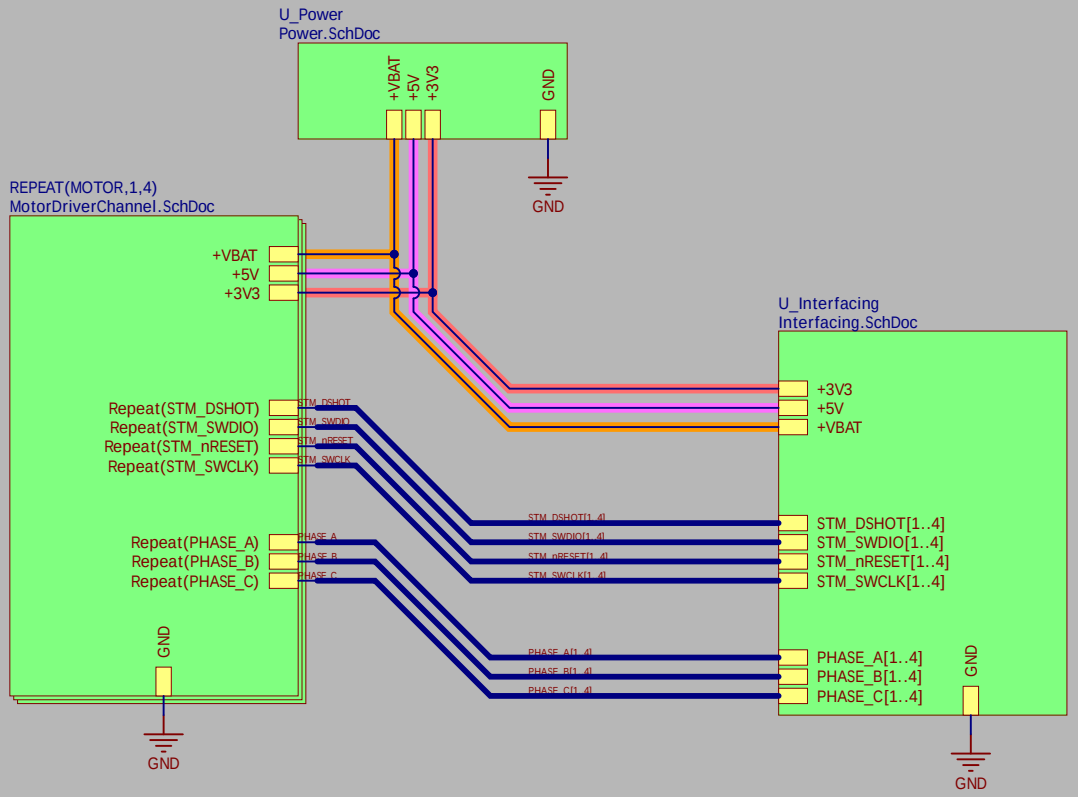
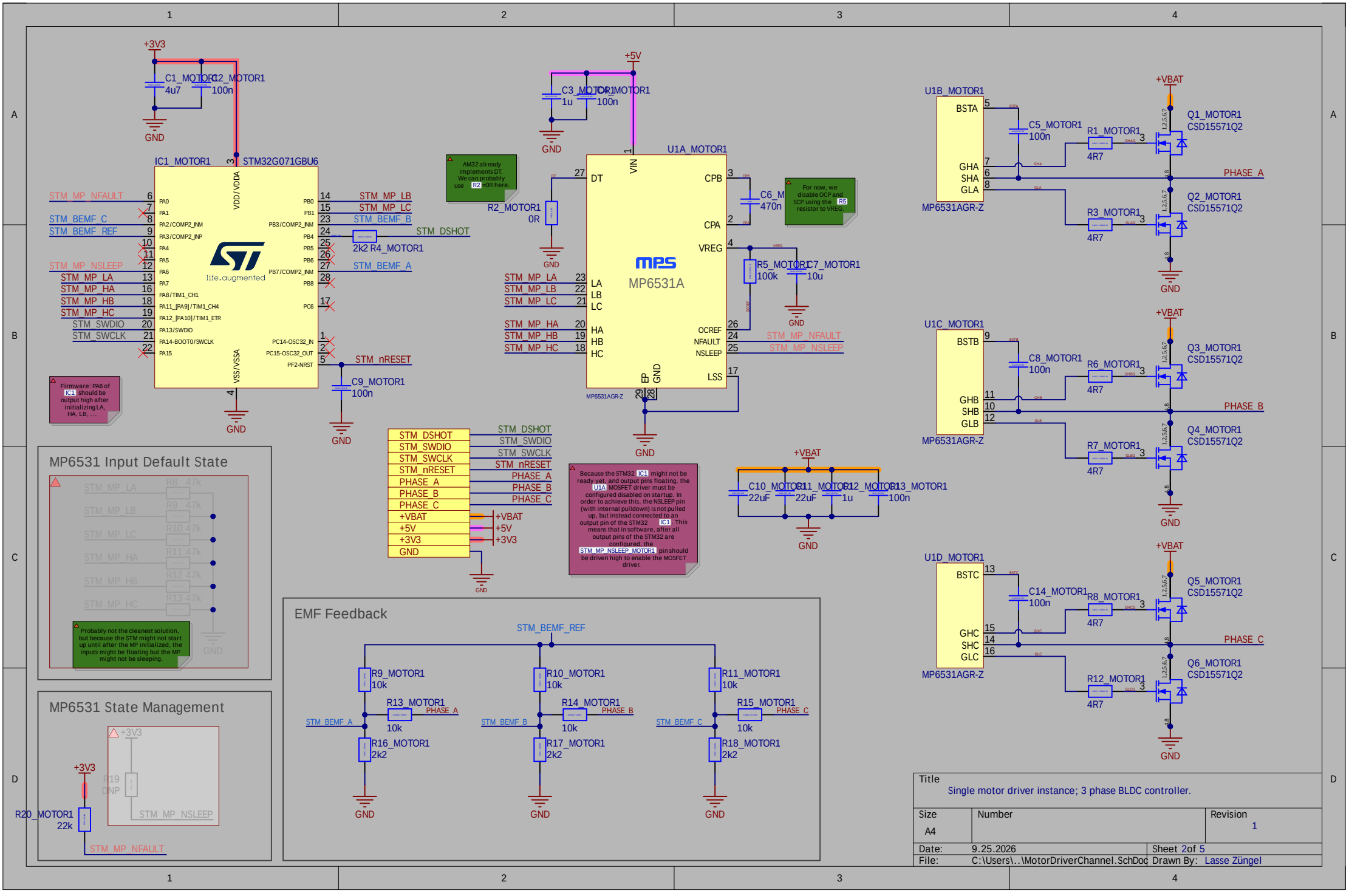
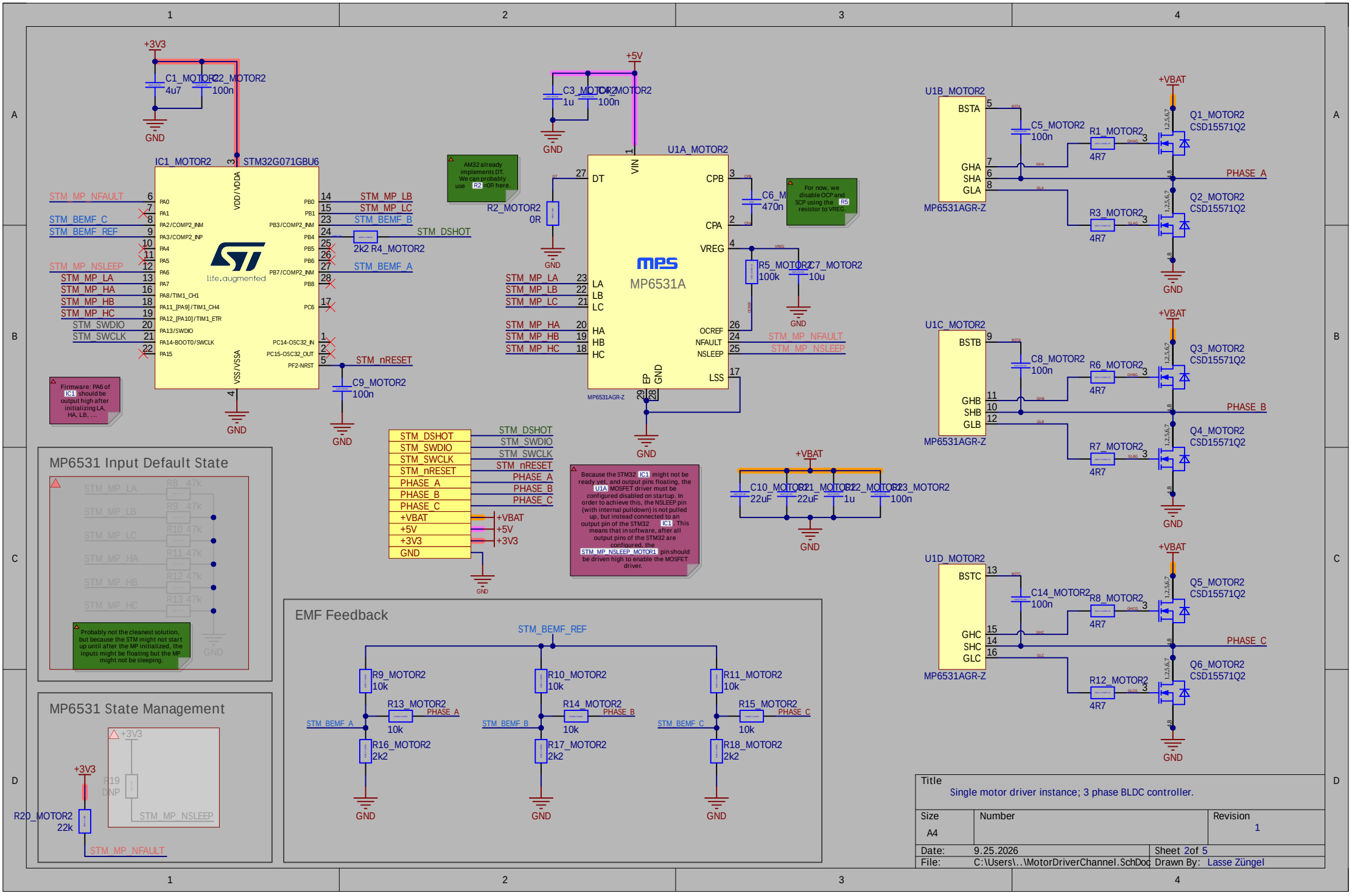
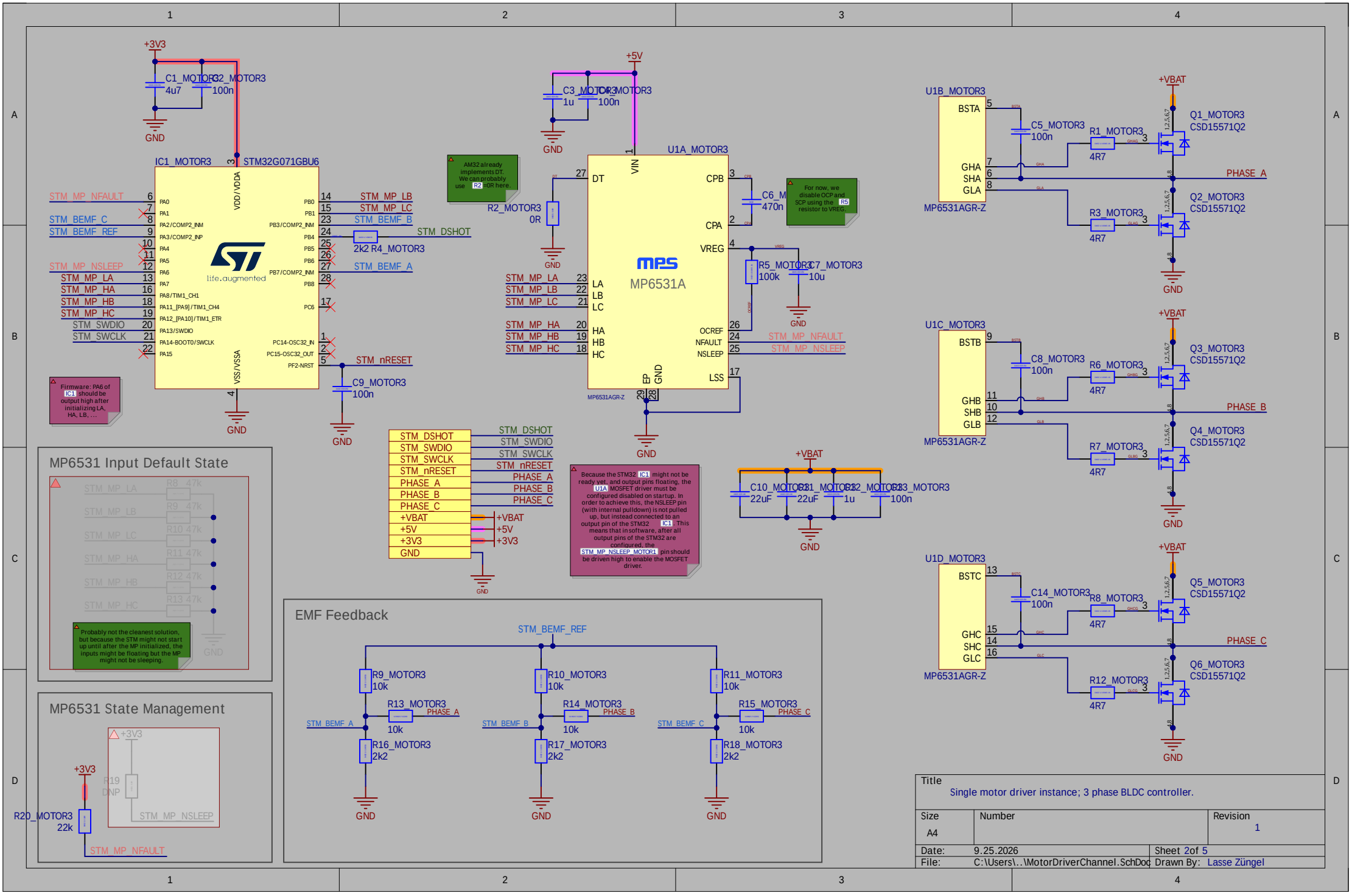
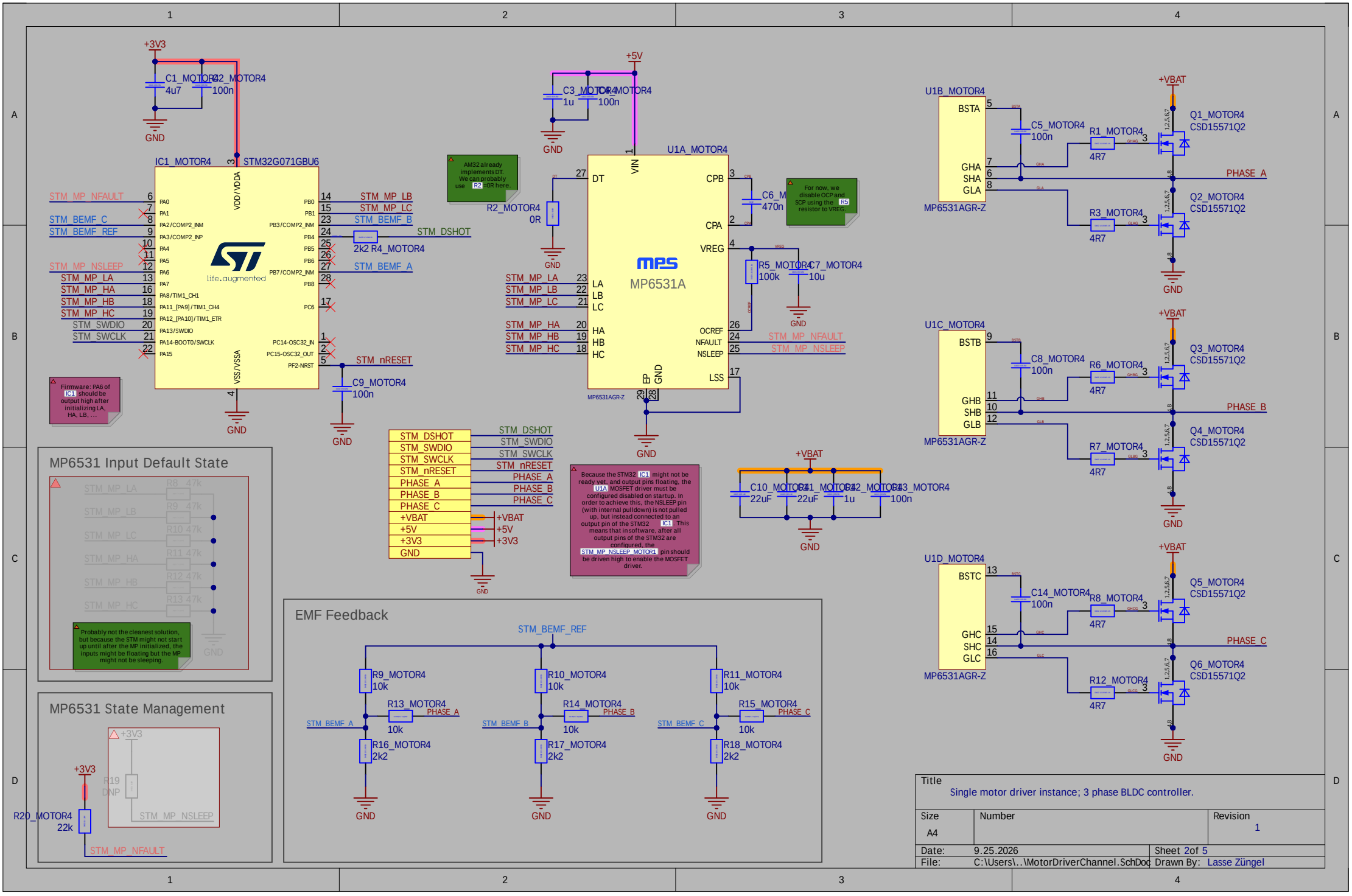


Title: Main page; instantiates the motor drivers, interface, and power.		
Size: A4	Number:	Revision: 1
Date: 9.25.2026		Sheet 1of5
File: C:\Users\..\Main.SchDoc		Drawn By: Lasse Züngel

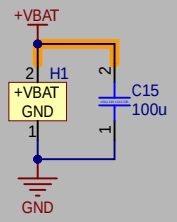

+3V3
C1_MOTOR1
4u7
C2_MOTOR1
100n
GND
IC1_MOTOR1
STM32G071GBU6
VDD/VDDA
VSS/VSSA
life.augmented
STM_MP_NFAULT
STM_BEMF_C
STM_BEMF_REF
STM_MP_NSLEEP
STM_MP_LA
STM_MP_HA
STM_MP_HB
STM_MP_HC
STM_SWDIO
STM_SWCLK
STM_MP_LB
STM_MP_LC
STM_BEMF_B
STM_DSHOT
2k2 R4_MOTOR1
STM_BEMF_A
STM_nRESET
C9_MOTOR1
100n
Firmware: PA6 of IC1 should be output high after initializing LA, HA, LB, ...
+5V
C3_MOTOR1
1u
C4_MOTOR1
100n
U1A_MOTOR1
MP6531A
MP6531AGR-Z
VIN
DT
CPB
CPA
VREG
OCREF
NFAULT
NSLEEP
LSS
EP
GND
AM32 already implements DT. We can probably use R2 = 0R here.
R2_MOTOR1
0R
C6_M
470n
For now, we disable OCP and SCP using the R5 resistor to VREG
R5_MOTOR1
100k
C7_MOTOR1
10u
STM_DSHOT
STM_SWDIO
STM_SWCLK
STM_nRESET
PHASE_A
PHASE_B
PHASE_C
+VBAT
+5V
+3V3
GND
Because the STM32 IC1 might not be ready yet, and output pins floating, the U1A MOSFET driver must be configured disabled on startup. In order to achieve this, the NSLEEP pin (with internal pulldown) is not pulled up, but instead connected to an output pin of the STM32 IC1. This means that in software, after all output pins of the STM32 are configured, the STM_MP_NSLEEP_MOTOR1 pin should be driven high to enable the MOSFET driver.
+VBAT
C10_MOTOR1
22uF
C11_MOTOR1
22uF
C12_MOTOR1
1u
C13_MOTOR1
100n
U1B_MOTOR1
BSTA
GHA
SHA
GLA
MP6531AGR-Z
C5_MOTOR1
100n
R1_MOTOR1
4R7
Q1_MOTOR1
CSD15571Q2
PHASE_A
R3_MOTOR1
4R7
Q2_MOTOR1
CSD15571Q2
U1C_MOTOR1
BSTB
GHB
SHB
GLB
C8_MOTOR1
100n
R6_MOTOR1
4R7
Q3_MOTOR1
CSD15571Q2
PHASE_B
R7_MOTOR1
4R7
Q4_MOTOR1
CSD15571Q2
U1D_MOTOR1
BSTC
GHC
SHC
GLC
C14_MOTOR1
100n
R8_MOTOR1
4R7
Q5_MOTOR1
CSD15571Q2
PHASE_C
R12_MOTOR1
4R7
Q6_MOTOR1
CSD15571Q2
MP6531 Input Default State
STM_MP_LA
STM_MP_LB
STM_MP_LC
STM_MP_HA
STM_MP_HB
STM_MP_HC
R8 47k
R9 47k
R10 47k
R11 47k
R12 47k
R13 47k
Probably not the cleanest solution, but because the STM might not start up until after the MP initialized, the inputs might be floating but the MP might not be sleeping.
MP6531 State Management
+3V3
R19
DNP
STM_MP_NSLEEP
R20_MOTOR1
22k
STM_MP_NFAULT
EMF Feedback
STM_BEMF_REF
R9_MOTOR1
10k
R13_MOTOR1
PHASE_A
10k
R16_MOTOR1
2k2
STM_BEMF_A
R10_MOTOR1
10k
R14_MOTOR1
PHASE_B
10k
R17_MOTOR1
2k2
STM_BEMF_B
R11_MOTOR1
10k
R15_MOTOR1
PHASE_C
10k
R18_MOTOR1
2k2
STM_BEMF_C
Title
Single motor driver instance; 3 phase BLDC controller.
Size
A4
Number
Revision
1
Date:
9.25.2026
Sheet 2of 5
File:
C:\Users\..\MotorDriverChannel.SchDoc
Drawn By:
Lasse Züngel

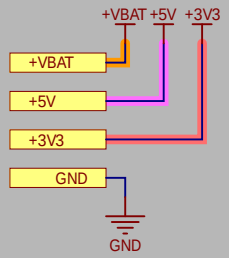

+3V3
C1_MOTOR2
4u7
C2_MOTOR2
100n
GND
IC1_MOTOR2
STM32G071GBU6
VDD/VDDA
VSS/VSSA
life.augmented
STM_MP_NFAULT
STM_BEMF_C
STM_BEMF_REF
STM_MP_NSLEEP
STM_MP_LA
STM_MP_HA
STM_MP_HB
STM_MP_HC
STM_SWDIO
STM_SWCLK
STM_MP_LB
STM_MP_LC
STM_BEMF_B
STM_DSHOT
2k2 R4_MOTOR2
STM_BEMF_A
STM_nRESET
C9_MOTOR2
100n
Firmware: PA6 of IC1 should be output high after initializing LA, HA, LB, ...
+5V
C3_MOTOR2
1u
C4_MOTOR2
100n
AM32 already implements DT. We can probably use R2 = 0R here.
R2_MOTOR2
0R
U1A_MOTOR2
MP6531A
MP6531AGR-Z
VIN
DT
CPB
CPA
VREG
LA
LB
LC
HA
HB
HC
OCREF
NFAULT
NSLEEP
LSS
EP
GND
C6_M
470n
For now, we disable OCP and SCP using the R5 resistor to VREG
R5_MOTOR2
100k
C7_MOTOR2
10u
STM_MP_NFAULT
STM_MP_NSLEEP
STM_DSHOT
STM_SWDIO
STM_SWCLK
STM_nRESET
PHASE_A
PHASE_B
PHASE_C
+VBAT
+5V
+3V3
GND
Because the STM32 IC1 might not be ready yet, and output pins floating, the U1A MOSFET driver must be configured disabled on startup. In order to achieve this, the NSLEEP pin (with internal pulldown) is not pulled up, but instead connected to an output pin of the STM32 IC1. This means that in software, after all output pins of the STM32 are configured, the STM_MP_NSLEEP_MOTOR1 pin should be driven high to enable the MOSFET driver.
+VBAT
C10_MOTOR2
22uF
C11_MOTOR2
22uF
C12_MOTOR2
1u
C13_MOTOR2
100n
GND
U1B_MOTOR2
U1C_MOTOR2
U1D_MOTOR2
MP6531AGR-Z
BSTA
GHA
SHA
GLA
BSTB
GHB
SHB
GLB
BSTC
GHC
SHC
GLC
C5_MOTOR2
100n
C8_MOTOR2
100n
C14_MOTOR2
100n
R1_MOTOR2
R3_MOTOR2
R6_MOTOR2
R7_MOTOR2
R8_MOTOR2
R12_MOTOR2
4R7
Q1_MOTOR2
CSD15571Q2
Q2_MOTOR2
CSD15571Q2
Q3_MOTOR2
CSD15571Q2
Q4_MOTOR2
CSD15571Q2
Q5_MOTOR2
CSD15571Q2
Q6_MOTOR2
CSD15571Q2
PHASE_A
PHASE_B
PHASE_C
MP6531 Input Default State
STM_MP_LA
STM_MP_LB
STM_MP_LC
STM_MP_HA
STM_MP_HB
STM_MP_HC
R8 47k
R9 47k
R10 47k
R11 47k
R12 47k
R13 47k
Probably not the cleanest solution, but because the STM might not start up until after the MP initialized, the inputs might be floating but the MP might not be sleeping.
MP6531 State Management
+3V3
R19
DNP
STM_MP_NSLEEP
R20_MOTOR2
22k
STM_MP_NFAULT
EMF Feedback
STM_BEMF_REF
R9_MOTOR2
10k
R10_MOTOR2
10k
R11_MOTOR2
10k
R13_MOTOR2
10k
R14_MOTOR2
10k
R15_MOTOR2
10k
R16_MOTOR2
2k2
R17_MOTOR2
2k2
R18_MOTOR2
2k2
STM_BEMF_A
STM_BEMF_B
STM_BEMF_C
PHASE_A
PHASE_B
PHASE_C
Title
Single motor driver instance; 3 phase BLDC controller.
Size
A4
Number
Revision
1
Date:
9.25.2026
Sheet 2of 5
File:
C:\Users\..\MotorDriverChannel.SchDoc
Drawn By:
Lasse Züngel


+3V3
C1_MOTOR3 4u7
C2_MOTOR3 100n
GND
IC1_MOTOR3
STM32G071GBU6
STM_MP_NFAULT
STM_BEMF_C
STM_BEMF_REF
STM_MP_NSLEEP
STM_MP_LA
STM_MP_HA
STM_MP_HB
STM_MP_HC
STM_SWDIO
STM_SWCLK
STM_MP_LB
STM_MP_LC
STM_BEMF_B
STM_DSHOT
2k2 R4_MOTOR3
STM_BEMF_A
STM_nRESET
C9_MOTOR3 100n
Firmware: PA6 of IC1 should be output high after initializing LA, HA, LB, ...
+5V
C3_MOTOR3 1u
C4_MOTOR3 100n
U1A_MOTOR3
MP6531A
AM32 already implements DT. We can probably use R2 = 0R here.
R2_MOTOR3 0R
For now, we disable OCP and SCP using the R5 resistor to VREG.
C6_M 470n
R5_MOTOR3 100k
C7_MOTOR3 10u
STM_MP_NFAULT
STM_MP_NSLEEP
Because the STM32 IC1 might not be ready yet, and output pins floating, the U1A MOSFET driver must be configured disabled on startup. In order to achieve this, the NSLEEP pin (with internal pulldown) is not pulled up, but instead connected to an output pin of the STM32 IC1. This means that in software, after all output pins of the STM32 are configured, the STM_MP_NSLEEP_MOTOR1 pin should be driven high to enable the MOSFET driver.
+VBAT
C10_MOTOR3 22uF
C11_MOTOR3 22uF
C12_MOTOR3 1u
C13_MOTOR3 100n
U1B_MOTOR3
MP6531AGR-Z
C5_MOTOR3 100n
R1_MOTOR3 4R7
R3_MOTOR3 4R7
Q1_MOTOR3 CSD15571Q2
Q2_MOTOR3 CSD15571Q2
PHASE_A
U1C_MOTOR3
C8_MOTOR3 100n
R6_MOTOR3 4R7
R7_MOTOR3 4R7
Q3_MOTOR3 CSD15571Q2
Q4_MOTOR3 CSD15571Q2
PHASE_B
U1D_MOTOR3
C14_MOTOR3 100n
R8_MOTOR3 4R7
R12_MOTOR3 4R7
Q5_MOTOR3 CSD15571Q2
Q6_MOTOR3 CSD15571Q2
PHASE_C
MP6531 Input Default State
R8 47k
R9 47k
R10 47k
R11 47k
R12 47k
R13 47k
Probably not the cleanest solution, but because the STM might not start up until after the MP initialized, the inputs might be floating but the MP might not be sleeping.
MP6531 State Management
R19 DNP
R20_MOTOR3 22k
EMF Feedback
R9_MOTOR3 10k
R10_MOTOR3 10k
R11_MOTOR3 10k
R13_MOTOR3 10k
R14_MOTOR3 10k
R15_MOTOR3 10k
R16_MOTOR3 2k2
R17_MOTOR3 2k2
R18_MOTOR3 2k2
Title
Single motor driver instance; 3 phase BLDC controller.
Size A4
Number
Revision 1
Date: 9.25.2026
Sheet 2of 5
File: C:\Users\..\MotorDriverChannel.SchDoc
Drawn By: Lasse Züngel

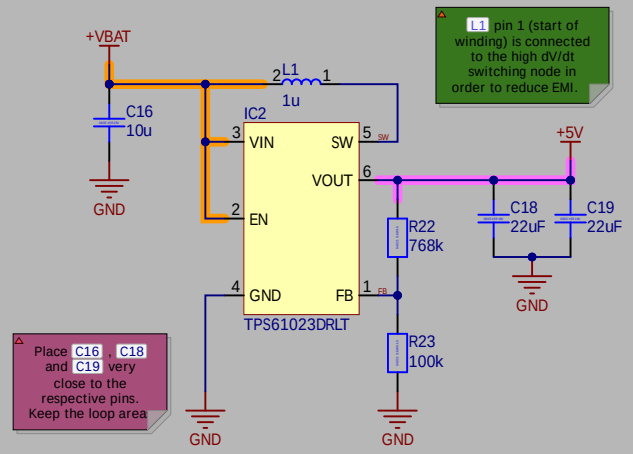

+3V3
C1_MOTOR4
4u7
C2_MOTOR4
100n
GND
IC1_MOTOR4
STM32G071GBU6
VDD/VDDA
VSS/VSSA
life.augmented
STM_MP_NFAULT
STM_BEMF_C
STM_BEMF_REF
STM_MP_NSLEEP
STM_MP_LA
STM_MP_HA
STM_MP_HB
STM_MP_HC
STM_SWDIO
STM_SWCLK
STM_MP_LB
STM_MP_LC
STM_BEMF_B
STM_DSHOT
2k2 R4_MOTOR4
STM_BEMF_A
STM_nRESET
C9_MOTOR4
100n
Firmware: PA6 of IC1 should be output high after initializing LA, HA, LB, ...
+5V
C3_MOTOR4
1u
C4_MOTOR4
100n
U1A_MOTOR4
MP6531A
MP6531AGR-Z
VIN
DT
CPB
CPA
VREG
LA
LB
LC
HA
HB
HC
OCREF
NFAULT
NSLEEP
LSS
EP
GND
AM32 already implements DT. We can probably use R2 = 0R here.
R2_MOTOR4
0R
C6_M
470n
For now, we disable OCP and SCP using the R5 resistor to VREG
R5_MOTOR4
100k
C7_MOTOR4
10u
STM_MP_NFAULT
STM_MP_NSLEEP
STM_DSHOT
STM_SWDIO
STM_SWCLK
STM_nRESET
PHASE_A
PHASE_B
PHASE_C
+VBAT
+5V
+3V3
GND
Because the STM32 IC1 might not be ready yet, and output pins floating, the U1A MOSFET driver must be configured disabled on startup. In order to achieve this, the NSLEEP pin (with internal pulldown) is not pulled up, but instead connected to an output pin of the STM32 IC1. This means that in software, after all output pins of the STM32 are configured, the STM_MP_NSLEEP_MOTOR1 pin should be driven high to enable the MOSFET driver.
+VBAT
C10_MOTOR4
22uF
C11_MOTOR4
22uF
C12_MOTOR4
1u
C13_MOTOR4
100n
GND
MP6531 Input Default State
STM_MP_LA
STM_MP_LB
STM_MP_LC
STM_MP_HA
STM_MP_HB
STM_MP_HC
R8 47k
R9 47k
R10 47k
R11 47k
R12 47k
R13 47k
Probably not the cleanest solution, but because the STM might not start up until after the MP initialized, the inputs might be floating but the MP might not be sleeping.
MP6531 State Management
+3V3
R19
DNP
STM_MP_NSLEEP
R20_MOTOR4
22k
STM_MP_NFAULT
EMF Feedback
STM_BEMF_REF
R9_MOTOR4
10k
R10_MOTOR4
10k
R11_MOTOR4
10k
R13_MOTOR4
10k
PHASE_A
R14_MOTOR4
10k
PHASE_B
R15_MOTOR4
10k
PHASE_C
STM_BEMF_A
STM_BEMF_B
STM_BEMF_C
R16_MOTOR4
2k2
R17_MOTOR4
2k2
R18_MOTOR4
2k2
U1B_MOTOR4
U1C_MOTOR4
U1D_MOTOR4
MP6531AGR-Z
BSTA
GHA
SHA
GLA
BSTB
GHB
SHB
GLB
BSTC
GHC
SHC
GLC
C5_MOTOR4
100n
C8_MOTOR4
100n
C14_MOTOR4
100n
R1_MOTOR4
4R7
R2_MOTOR4
4R7
R3_MOTOR4
4R7
R6_MOTOR4
4R7
R7_MOTOR4
4R7
R8_MOTOR4
4R7
R12_MOTOR4
4R7
Q1_MOTOR4
CSD15571Q2
Q2_MOTOR4
CSD15571Q2
Q3_MOTOR4
CSD15571Q2
Q4_MOTOR4
CSD15571Q2
Q5_MOTOR4
CSD15571Q2
Q6_MOTOR4
CSD15571Q2
PHASE_A
PHASE_B
PHASE_C
Title
Single motor driver instance; 3 phase BLDC controller.
Size
A4
Number
Revision
1
Date:
9.25.2026
Sheet 2of 5
File:
C:\Users\..\MotorDriverChannel.SchDoc
Drawn By:
Lasse Züngel

Battery Connection

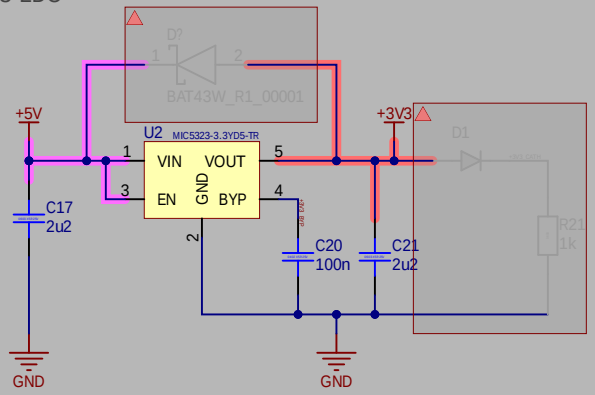




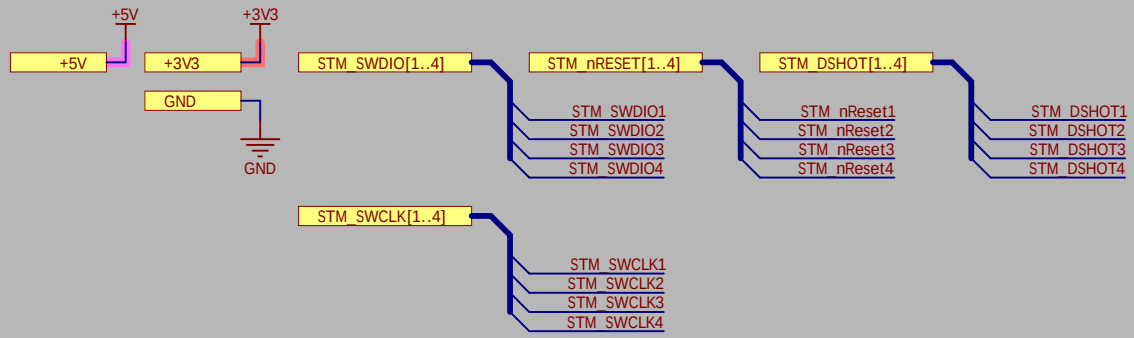
5V Boost Converter from Battery Voltage



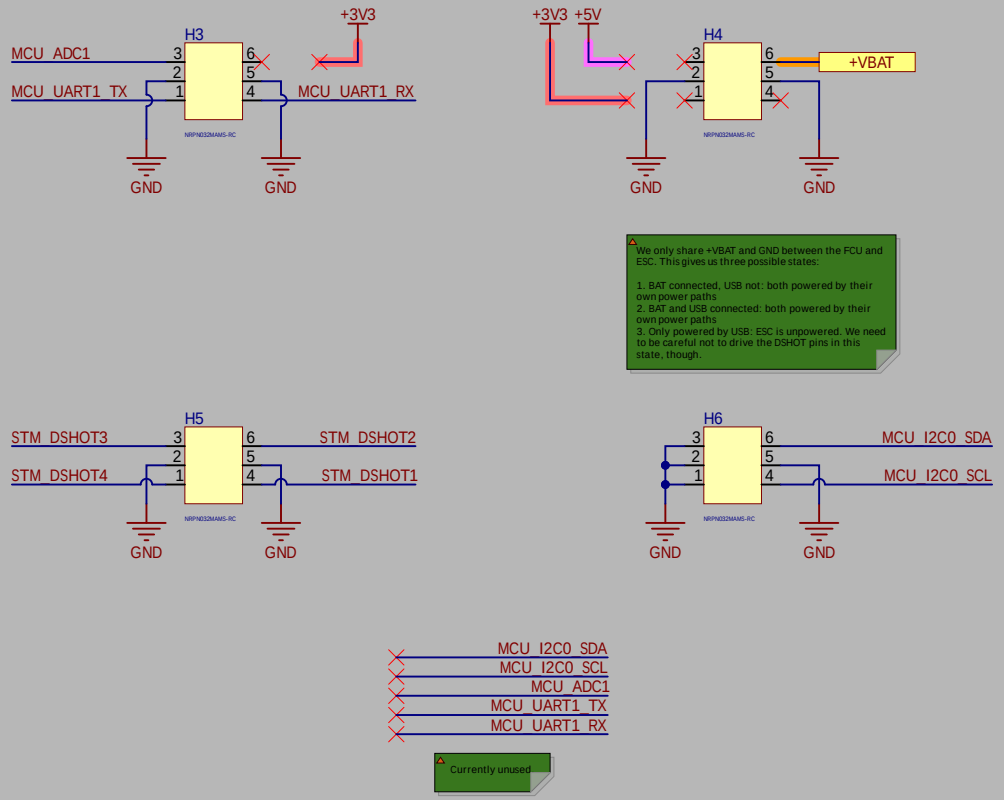
3V3 LDO



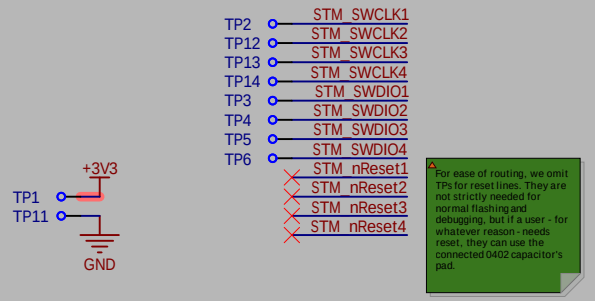
Title: Power supply, power management		
Size: A4	Number:	Revision: 1
Date: 9.25.2026		Sheet 3of5
File: C:\Users\..\Power.SchDoc		Drawn By: Lasse Züngel



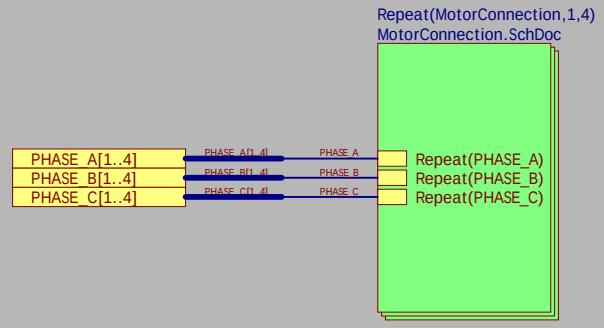
Four Main Headers, Connection to FCU



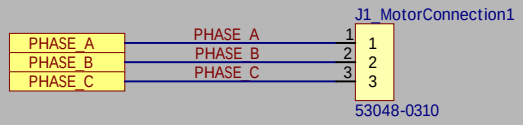
Pogo Connector for Bootloader Flashing



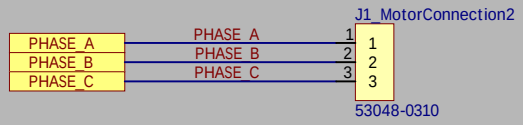
One Motor Connector (3 phases) per Motor



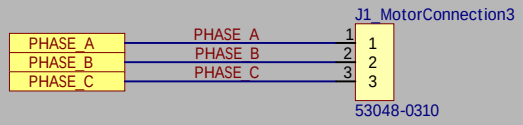
Title: Connections etc.		
Size: A4	Number:	Revision: 1
Date: 9.25.2026		Sheet 4of5
File: C:\Users\..\Interfacing.SchDoc		Drawn By: Lasse Züngel



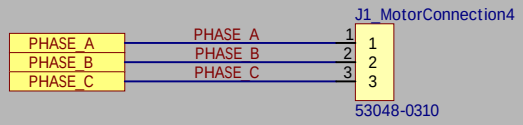
Title: Motor connections, per motor.		
Size: A4	Number:	Revision: 1
Date: 9.25.2026		Sheet 5 of 5
File: C:\Users\..\MotorConnection.SchDoc		Drawn By: Lasse Züngel



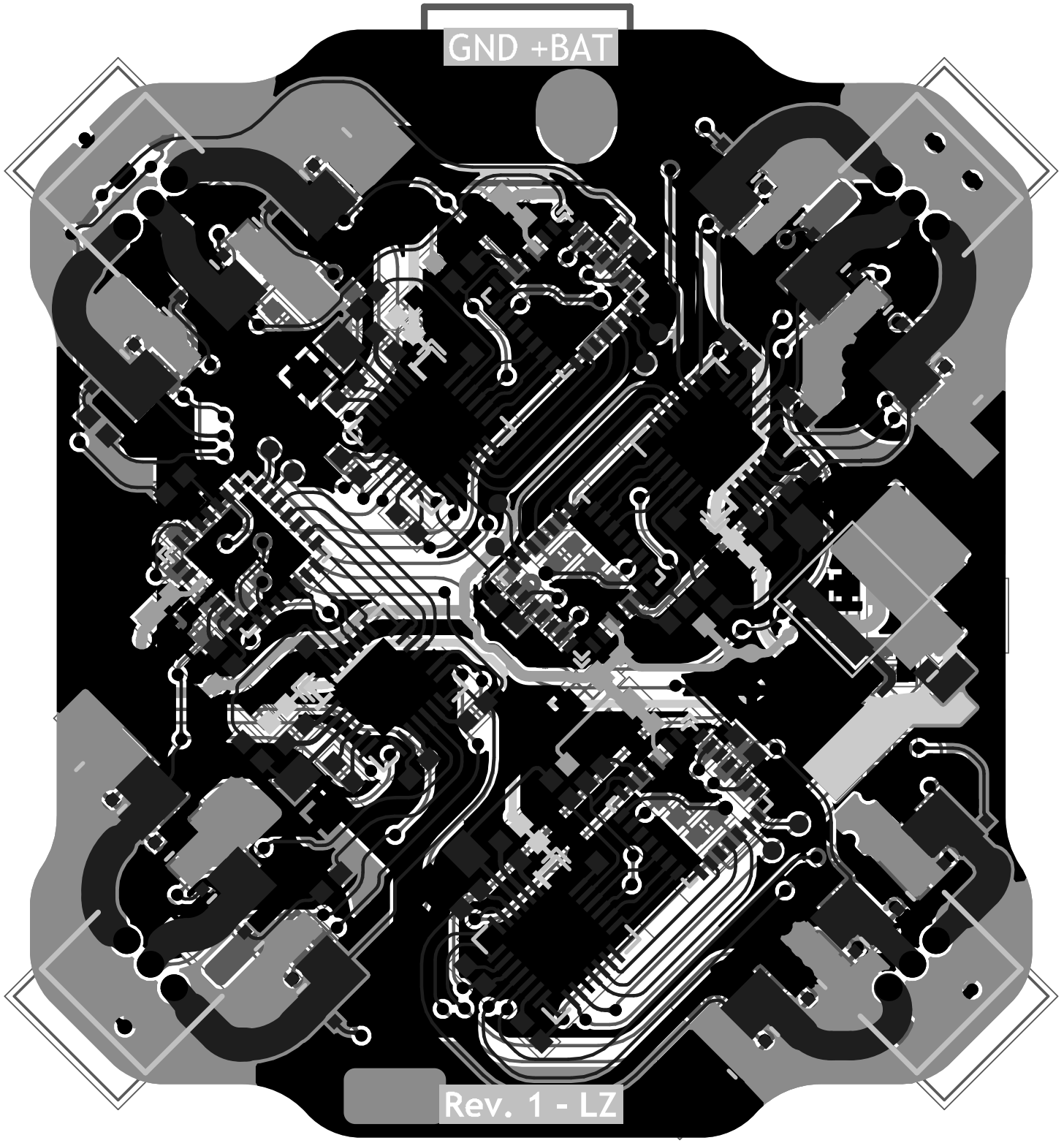
Title: Motor connections, per motor.		
Size: A4	Number:	Revision: 1
Date: 9.25.2026		Sheet 5 of 5
File: C:\Users\..\MotorConnection.SchDoc		Drawn By: Lasse Züngel



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